

set up curved TBL simulation in neko

extract k , ω and U at x position $Re_{thate} = 1000$

set up OpenFOAM simulation with k , ω and U as inflow

OPTIMIZATION

set new boundary points q
(Bayesian Optimization)

run OpenFOAM simulation

compute β and residual

update XX

extract U_x and U_y at $y_n = 24.5$ (neko upper wall)

import velocity profiles in neko simulation

run neko simulation

END